

- 1 (a) power = energy / time or work done / time B1
 force: kg m s^{-2} (including from mg in mgh or Fv)
 or kinetic energy ($\frac{1}{2}mv^2$): $\text{kg (m s}^{-1})^2$ B1
 (distance: m and (time) $^{-1}$: s^{-1}) and hence power: $\text{kg m s}^{-2} \text{ m s}^{-1} = \text{kg m}^2 \text{ s}^{-3}$ B1 [3]
- (b) Q/t : $\text{kg m}^2 \text{ s}^{-3}$ C1
 A : m^2 and x : m and T : K C1
 correct substitution into $C = (Qx) / tAT$ or equivalent, or with cancellation C1
 units of C : $\text{kg m s}^{-3} \text{ K}^{-1}$ A1 [4]